Problem I

Problem II

Problem III

Problem IV

Let $(X, \mathcal{T}_X)$ be a nonempty topological space. Consider $X^{\mathbb{N}}$ equipped with the product topology. For any $N \in \mathbb{N}$, we will also consider X^{N+1} equipped with the product topology. For any $M \leq N$, we introduce the maps

$$\begin{aligned} \Pi^X_{N,M} \colon X^{N+1} &\to X^{M+1} \\ (x_0, \dots, x_N) &\mapsto (x_0, \dots, x_M) \end{aligned}$$

and

$$\begin{aligned} \pi^X_{N,M} \colon X^{N+1} &\to X \\ (x_0, \dots, x_N) &\to X_M. \end{aligned}$$

We will also consider their $N = \infty$ analogues; for any $M \in \mathbb{N}$, we let

$$\begin{aligned} \Pi^X_{\infty,M} \colon X^{\mathbb{N}} &\to X^{M+1} \\ (x_0, x_1, \dots) &\to (x_0, \dots, x_M) \\ \pi^X_{\infty,M} \colon X^{\mathbb{N}} &\to X \\ (x_0, x_1, \dots) &\to x_M. \end{aligned}$$

All these maps are continuous by the definition of the product topology. We will call a sequence $(\mu_N)_{N \geq 0}$ with $\mu_N \in \mathrm{Prob}\left(X^{N+1}\right)$ *coherent* if for all $M, N \in \mathbb{N}$ with $M \leq N$, we have that $\left(\Pi^X_{N,M}\right)_* \mu_N = \mu_M$.

We will say that $(X, \mathcal{T}_X)$ satisfies the Daniell–Kolmogorov extension theorem (KET) if for all coherent sequences of probability measures $(\mu_N)_N$, there is a unique $\nu \in \mathrm{Prob}\left(X^{\mathbb{N}}\right)$ such that for all $N \geq 0$, one has $\left(\Pi^X_{\infty,N}\right)_* \nu = \mu_N$.

Problem (Problem IV.1)**:** If X and Y are continuous spaces, and $\theta \colon X \to Y$ is a map between them, then for any $N \geq 0$ we let

$$\begin{aligned} \theta^{(N)} \colon X^{N+1} &\to Y^{N+1} \\ (x_0, \dots, x_N) &\to (\theta(x_0), \dots, \theta(x_N)), \end{aligned}$$

and analogously define the infinite analogue $\theta^{(\infty)} \colon X^{\mathbb{N}} \to Y^{\mathbb{N}}$. Suppose θ is a homeomorphism. Show that X satisfies the KET if and only if Y does.

Solution: Suppose X satisfies the KET, and let $(\mu_N)_N$ be a coherent sequence of probability measures on Y. We will let $\Phi = \theta^{-1}$, which is a homeomorphism from Y to X. Consider the sequence of pushforwards $\left(\Phi^{(N)}_* \mu_N\right)_N$. We claim that this sequence is coherent with respect to X. This amounts to showing that for any $M \leq N$, we have that $\left(\Pi^X_{N,M}\right)_* \left(\Phi^{(N)}_* \mu_N\right) = \Phi^{(M)}_* \mu_M$.

Let $E = E_0 \times \cdots \times E_M$ be a Borel cylinder set in X^{M+1}, and let $F_i = \theta(E_i)$. Then, we have that

$$\begin{aligned} \Phi^{(M)}_* \mu_M (E_0 \times \cdots \times E_M) &= \mu_M\left(\theta^{(M)}(E_0 \times \cdots \times E_M)\right) \\ &= \mu_M(\theta(E_0) \times \cdots \times \theta(E_M)) \end{aligned}$$

$$= \mu_M(F_1 \times \cdots \times F_M),$$

while

$$\begin{aligned}
\left(\Phi_{N,M}^X\right)_*\left(\Phi_*^{(N)}\mu_N\right)(E_0 \times \cdots \times E_M) &= \mu_N\left(\theta^{(N)}\left(E_0 \times \cdots \times E_M \times X^{N-M}\right)\right) \\
&= \mu_N\left(\theta(E_0) \times \cdots \times \theta(E_M) \times Y^{N-M}\right) \\
&= \mu_N\left(F_0 \times \cdots \times F_M \times Y^{N-M}\right) \\
&= \mu_N\left(\left(\Pi_{N,M}^Y\right)^{-1}(F_0 \times \cdots \times F_M)\right) \\
&= \mu_M(F_0 \times \cdots \times F_M),
\end{aligned}$$

where the latter equality emerges from the fact that the $(\mu_N)_N$ are coherent on Y. Since X satisfies the KET and $(\Phi_*\mu_N)_N$ is coherent on X, there is a unique probability measure ν on $X^{\mathbb{N}}$ such that $\left(\Pi_{\infty,M}^X\right)_*\nu = \Phi_*^{(M)}\mu_M$. We claim that $\theta_*^{(\infty)}\nu$ is the desired probability measure on $Y^{\mathbb{N}}$. This amounts to verifying that $\left(\Pi_{\infty,M}^Y\right)_*\theta_*^{(\infty)}\nu = \mu_M$ on Borel cylinder sets in $Y^{\mathbb{N}}$, as the cylinder sets generate the Borel σ-algebra on $Y^{\mathbb{N}}$ and any measure is uniquely determined by its value on the cylinder sets by the Carathéodory extension theorem.

Let $F_0 \times \cdots \times F_M \times Y^{\mathbb{N}}$ be a Borel cylinder set in $Y^{\mathbb{N}}$. Write $E_i = \Phi(F_i)$. Then, we have that

$$\begin{aligned}
\left(\Pi_{\infty,M}^Y\right)_*\left(\theta_*^{(\infty)}\nu\right)\left(F_0 \times \cdots \times F_M \times Y^{\mathbb{N}}\right) &= \left(\Pi_{\infty,M}^Y \circ \theta^{(\infty)}\right)_*\nu\left(F_0 \times \cdots \times F_M \times Y^{\mathbb{N}}\right) \\
&= \left(\theta^{(M)} \circ \Pi_{\infty,M}^X\right)_*\nu\left(F_0 \times \cdots \times F_M \times Y^{\mathbb{N}}\right) \\
&= \left(\theta^{(M)}\right)_*\left(\Pi_{\infty,M}^X\right)_*\nu\left(F_0 \times \cdots \times F_M \times Y^{\mathbb{N}}\right) \\
&= \theta_*^{(N)}\Phi_*^{(N)}\mu_M(F_0 \times \cdots \times F_M) \\
&= \left(\theta^{(N)} \circ \Phi^{(N)}\right)_*\mu_M(F_0 \times \cdots \times F_M) \\
&= \mu_M(F_0 \times \cdots \times F_M).
\end{aligned}$$

The reverse direction follows from swapping all instances of θ with instances of Φ and all instances of X with instances of Y.